



Degree Project in Decision and Control Systems

Second cycle, 30 credits

Patient-Specific Parameter Estimation in Mechanistic Insulin-Glucose Models

Degree Project 2026

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Master's Programme, Information and Network Engineering, 120 credits
Date: May 29, 2026

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Swedish title: Detta är den svenska översättningen av titeln
Swedish subtitle: Detta är den svenska översättningen av undertiteln

Abstract

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Keywords

Canvas Learning Management System, Docker containers, Performance tuning

Sammanfattning

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Nyckelord

Canvas Lärplattform, Dockerbehållare, Prestandajustering

Acknowledgments

Stockholm, May 2026

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Chapter 1

Introduction

Patients in the intensive care unit suffering from critical illnesses (e.g. acute myocardial infarction, sepsis, stroke) have been widely documented to undergo hyperglycaemia, even those without a prior diagnosis of diabetes [1]. Acute stress observed in critical illnesses often induces physiological changes, as the secretion of cortisol and catecholamines results in an inflammatory response associated with insulin-resistance [2] [3] [4] [5] [6] [7]. While there is inconclusive evidence pertaining to the causality of stress hyperglycaemia and adverse outcomes, the onset of mortality is far greater in patients with no reported history of hyperglycaemia, in contrast to patients with diabetes [1] [3]. Sustaining glucose concentration under 6.106 mmol/L through insulin infusion has demonstrated a reduction of mortality and morbidity, compared to less-intensive methods of glycaemic control [8], at the subsequent cost of an increase of reported incidences of hypoglycaemia [9]. Further review of intensive insulin therapy and tight glucose control has identified discrepancies in the administration of glycaemic control, mostly corresponding to the risk of moderate to severe hypoglycaemia and glycaemic variability in a critically ill patient is equally detrimental as stress hyperglycaemia, with an increase of mortality and health complications observed [9] [10] [11]. Clinicians aim to maintain blood glucose in a range of 6 mmol/L to 10 mmol/L to account for adverse outcomes in hyperglycaemia and hypoglycaemia. Traditional intensive insulin therapy struggles to fully capture the dynamic fluctuations of blood glucose in critically ill patients. Thus, model-based methods are required to achieve tight glycaemic control.

1.1 Background

The implementation of model-based tight glucose control methods relies on the derivation of insulin-glucose dynamics. Mechanistic insulin-glucose models have been developed with an extensive physiological foundation, and have been validated for gluco-regulation in clinical settings [12] [13] [14] [15] [16]. The insulin-glucose models derived are nonlinear, which requires more advanced numerical methods to properly estimate the current and future insulin-glucose dynamics in patients. In practice, the estimated dynamics from the insulin-glucose models contain process noise from unaccounted dynamics in the model formulation as well as noisy measurements. As a result, the insulin-glucose dynamics are not deterministic. To accommodate modeling of stochastic processes, statistical methods are needed to optimally estimate the dynamics in mechanistic insulin-glucose models.

1.2 Problem

1.2.1 Original problem and definition

There currently exist gaps in research for in-depth analysis of statistical modeling of insulin-glucose dynamics. Previous studies have employed Bayesian filtering methods for this task [17] [18], but there lacks necessary evaluation and motivation regarding the patient cohort and the filter design process. While patient models differ among institutions due to differing nutrition guidelines and stay times for patients, a systematic virtual patient cohort framework would greatly improve the simulation of diverse patient dynamics and experiment reproducibility, as real patient data are often limited in size and closed-source. Concurrently, concise parameterization and optimization of Bayesian filters will further improve glycaemic control in critically-ill patients.

1.2.2 Scientific and engineering issues

To model the behavior of dynamical systems, experience of statistical modeling is often required to construct the assumptions of the system parameters. In the case of insulin-glucose dynamics, there are several unknown patient-specific parameters, and each has its respective role in modeling gluco-regulation. As a result, on top of estimating nonlinear and

time-varying dynamics, the models must be robust to misalignments of the assigned patient-specific parameters, which poses an ongoing challenge.

1.3 Purpose

Mechanistic insulin-glucose models are of great benefit to critically ill patients and clinicians, as modeling gluco-regulation allows for necessary glycaemic adjustments to avoid the onset of adverse outcomes. In practice, clinicians only have the ability to base insulin resistance on blood glucose measurements, and thus, state estimators models fill in the gap as a tool for insulin-glucose dynamics estimation. Improvements in the estimation of insulin-glucose dynamics can better capture the evolution of the health of the patient in the intensive care unit to prevent mortality and morbidity.

1.4 Goals

The overall goals of this project has been divided into the following sub-goals:

1. Design a virtual patient cohort
2. Compare performance of varying filtering configurations in estimation of insulin-glucose dynamics
3. Extrapolate the formulation of insulin-glucose estimators to estimate unknown patient-specific parameters

1.5 Research Methodology

The modeling of the insulin-glucose dynamics found in critically ill patients will be implemented on a virtual patient cohort. Using virtual patients allows the researcher to manually control the patient-specific parameters, which directly improves the evaluation of the outcome of the insulin-glucose control. Previous work will serve as the benchmark to compare how the parameterization of our method performs in estimating insulin-glucose dynamics of virtual patients. The chosen filter design will then be used to estimate additional dynamics not mathematically represented in insulin-glucose models. Fitting metrics and computational complexity of the estimators will be utilized to draw inferences about the performance.

1.6 Delimitations

This study will consist of a virtual patient cohort. Using virtual patients is important for maintaining internal validity in the experimental process, as it verifies the observed outcome of the experiment is due to the manipulation of patient-specific parameters rather than uncontrolled variables. The expense of using virtual patients can be identified in the reduced generalizability of this experiment, since virtual patients are models. While this is a limitation, this study will focus on the foundation of the modeling of insulin-glucose dynamics, and future work can replicate the study with real clinical data.

1.7 Structure of the thesis

Chapter 2 will introduce the existing modeling of insulin-glucose dynamics used, followed by an overview of discretization, state estimation, and related work in parameter estimation. In Chapter 3, the proposed research process used to answer the research questions of this study will be documented and motivated. The results will be presented in Chapter 4. The analysis of the results, key findings, and future work of the experiment will be summarized in Chapter 6.

Chapter 2

Background

The Intensive Control Insulin-Nutrition-Glucose model presented by Lin et al. [14] contains the insulin-glucose dynamics in critically ill patients. The model is constructed as a fusion of two existing models validated for patients in the intensive care unit [12] [13], containing the following ensemble of differential equations

$$\dot{G} = -p_G G(t) - S_I(t) G(t) \frac{Q(t)}{1 + \alpha_G Q(t)} + \frac{P(t) + E_G - C_G}{V_G} \quad (2.1)$$

$$\dot{Q} = n_I(I(t) - Q(t)) - n_C \frac{Q(t)}{1 + \alpha_G Q(t)} \quad (2.2)$$

$$\begin{aligned} \dot{I} = & -n_K I(t) - n_L \frac{I(t)}{1 + \alpha_I I(t)} - n_I(I(t) - Q(t)) + \frac{u_{ex}(t)}{V_I} \\ & + \frac{(1 - x_L)u_{en}(t)}{V_I} \end{aligned} \quad (2.3)$$

$$\dot{P1} = -d_1 P1 + D(t) \quad (2.4)$$

$$\dot{P2} = -\min(d_2 P2, P_{max}) + d_1 P1 \quad (2.5)$$

$$u_{en}(t) = k_1 \exp\left(-\frac{k_2}{k_3} I(t)\right), \quad \text{when C-peptide data is not available} \quad (2.6)$$

$$P(t) = \min(d_2 P2, P_{max}) + P_N(t). \quad (2.7)$$

Equation (2.1) is the independent insulin glucose removal, with G being the total blood glucose concentration (mmol/L). The whole-body insulin sensitivity $S_I(t)$ gives insight into the metabolic condition of the patient and is a time-varying dynamic. The insulin pharamacodynamics are shown in Equations (2.2) and (2.3), with Q and I being the interstitial insulin concentration (mU/L) and the plasma insulin concentration (mU/L)

respectively. The exogenous insulin input is $u_{ex}(t)$ (mU/min), and the endogenous insulin production $u_{en}(t)$ (mU/min) can be measured from C-Peptide, but if measurements are not feasible, the kinetics can model as such in Equation (2.6). The gastric absorption of glucose is defined in Equations (2.4) to (2.7). The dynamics $P1$ and $P2$ describe the glucose in the stomach and the gut (mmol), with $P(t)$ representing the glucose appearance in the blood (mmol/min). Any enteral food intake given to the patient is accounted for by $D(t)$ (mmol/min), and additional parenteral glucose is described by $P_N(t)$ (mmol/min).

The Intensive Control Insulin-Nutrition-Glucose model contains of several fixed parameters detailed in Table 2.1. The majority of the patient model parameters are assigned as general population constants observed from clinical trials, and are validated in relevant insulin-glucose control parameter sensitivity studies [15] [16] [19]. The identification of the remaining constant parameters was estimated with a data-driven approach by fitting the patient parameters to the observed dynamics in 173 patients, and by assessing the impact of prediction errors [19]. Estimating the endogenous glucose removal p_G and the suppression of endogenous glucose produce E_G constants involved assigning all other patient parameters and the insulin sensitivity to constants, then starting a grid search that covers $p_G = 0.001, 0.002, \dots, 0.1(\text{min}^{-1})$ and $E_G = 0.0, 0.1, \dots, 3.5$ (mmol/min). The joint p_G and E_G value that achieved the lowest fitting and prediction error for blood glucose in Equation (2.1) was chosen. Ultimately, that was $p_G = 0.006(\text{min}^{-1})$ and $E_G = 1.16$, while both falling in the ranges observed physiologically. The glucose distribution rate between the stomach and the gut n_I , and cellular insulin clearance rate in the interstitium were estimated similarly by fitting to Q in the Equation (2.2), a grid search covers a range of $n_I = n_C = 0.0001, \dots, 0.02(\text{min}^{-1})$, with a value of $n_I = 0.003$ resulting in the best predictive performance.

It should be noted that in the Intensive Control Insulin-Nutrition-Glucose model, the nonlinear dynamics are defined in continuous time. Discretization of the dynamic system is necessary as in the following section, the estimation methods presented operate within discrete time noted as k . If we define the discrete time step as $t_k = t_{k-1} + dt$, an example dynamic $F(t)$ can be discretized according to the Euler method

$$\begin{aligned}\dot{F} &= \frac{F(t_k) - F(t_k - dt)}{dt} \\ F_k &= F_{k-1} + dt\dot{F}.\end{aligned}\tag{2.8}$$

The error of this numerical approximation of differential equations is bounded

Table 2.1: Constant parameters in the Intensive Control Insulin-Nutrition-Glucose model.

Constant Parameter	Notation	Value
Patient endogenous glucose removal	p_G (min^{-1})	0.006
Saturation of insulin-stimulated glucose removal	α_G (L/mU)	0.0154
Suppression of endogenous glucose production	E_G (mmol/min)	1.16
Central nervous system uptake	C_G (mmol/L)	0.3
Glucose distribution volume	V_G (L)	13.3
Glucose distribution rate between the stomach and the gut	n_I (min^{-1})	0.003
Cellular insulin clearance rate in the interstitium	n_C (min^{-1})	$= n_I$
Kidney clearance rate of insulin from plasma	n_K (min^{-1})	0.0542
Liver clearance rate of insulin from plasma	n_L (min^{-1})	0.01578
Saturation of plasma insulin disappearance	α_I (L/mU)	0.0017
Insulin distribution volume	V_I (L)	3.15
Fraction of endogenous insulin hepatic extraction	x_L	0.67
Maximal glucose out flux from the gut	P_{max} (mmol/min)	6.11
Base endogenous insulin production rate	k_1 (mU/min)	45.7
Transport rates between the stomach and the gut	d_1 (min^{-1})	0.0347
	d_2 (min^{-1})	0.0069
Exponential suppression constants	k_2	1.5
	k_3	1000

Parameters shaded in gray are derived from fitting methods, as the rest are assigned as population constants.

by the step size dt . As a result, larger step sizes will introduce greater approximation errors, thus, a relatively small step size is necessary to prevent errors from propagating in the estimates.

2.1 State Estimation

Within the scope of this paper, discrete-time dynamical systems will be modeled in the state-space representation. A state-space model consists of system variables that describe the state of the system, inputs, and measurements. The states are typically formulated as linear or nonlinear differential equations. A probabilistic representation of the discrete-time state-space can be represented as such

$$\begin{aligned} x_k &\sim p(x_k \mid x_{k-1}, u_{k-1}) \\ y_k &\sim p(y_k \mid x_k), \end{aligned} \tag{2.9}$$

where $p(x_k | x_{k-1}, u_k)$ and $p(y_k | x_k)$ are the stochastic discrete-time dynamic and measurement models, with $x_k \in \mathbb{R}^n$ representing the states, $u_k \in \mathbb{R}^p$ are the inputs, and $y_k \in \mathbb{R}^m$ are the measurements. The objective is to ultimately derive the distribution of the state x_k , given all the measurements up to the current sample $p(x_k | y_{1:k})$. In the case of assuming the state estimator is linear and Gaussian, the states and measurements can be written as the difference equations

$$\begin{aligned} x_k &= A_{k-1}x_{k-1} + B_{k-1}u_{k-1} + q_{k-1} \\ y_k &= Hx_k + r_k, \end{aligned} \quad (2.10)$$

where $q_{k-1} \sim N(0, Q_{k-1})$ is the process noise, $r_k \sim N(0, R_k)$ is the measurement noise, we can rewrite the Equation (2.9) as the following

$$\begin{aligned} p(x_k | x_{k-1}, u_{k-1}) &= N(x_k | A_{k-1}x_{k-1} + B_{k-1}u_{k-1}, Q_{k-1}) \\ p(y_k | x_k) &= N(y_k | H_k x_k, R_k). \end{aligned} \quad (2.11)$$

The closed-form solution of $p(x_k | y_{1:k})$ for linear and Gaussian dynamic and measurement functions is known as the Kalman filter, with a full derivation available from Särkkä and Svensson [20]. The filter contains two key parts: the *prediction step* and the *update step*.

1. Prediction step

$$\begin{aligned} m_k^- &= A_{k-1}x_{k-1} + B_{k-1}u_{k-1} \\ P_k^- &= A_{k-1}P_{k-1}A_{k-1}^T + Q_{k-1}. \end{aligned} \quad (2.12)$$

2. Update step

$$\begin{aligned} v_k &= y_k - H_k m_k^- \\ S_k &= H_k P_k^- H_k^T + R_k \\ K_k &= P_k^- H_k^T S_k^{-1} \\ m_k &= m_k^- + K_k v_k \\ P_k &= P_k^- - K_k S_k K_k^T. \end{aligned} \quad (2.13)$$

The Kalman filter is recursive, as the filter starts with an estimate of the initial distribution of the state m_0^- and the covariance P_0^- , and iterates sequentially over discrete time steps k . In practice, the Kalman filter is often a great starting point for an estimator because the assumptions of linearity and Gaussianity often hold in many contexts, but in the case of nonlinear dynamic and measurement models, other methods must be considered.

2.1.1 Extended Kalman filter

The extended Kalman filter can accommodate nonlinearities by approximating the dynamic and measurement models as a linear transformation of the states and inputs. The structure of the extended Kalman filter remains the same as the Equations (2.12) and (2.13), but changes are seen in the matrices of the dynamic and measurement models $f(\cdot)$ and $h(\cdot)$. Assuming that the dynamic and measurement models are Gaussian, the prediction and update steps [20] can be defined as

1. Prediction step

$$\begin{aligned} m_k^- &= f(m_{k-1}, u_{k-1}) \\ P_k^- &= F_x(m_{k-1})P_{k-1}F_x^T(m_{k-1}) + Q_{k-1} \end{aligned} \quad (2.14)$$

2. Update step

$$\begin{aligned} v_k &= y_k - h(m_k^-) \\ S_k &= H_x(m_k^-)P_k^-H_x^T(m_k^-) + R_k \\ K_k &= P_k^-H_k^T(m_k^-)S_k^{-1} \\ m_k &= m_k^- + K_kv_K \\ P_k &= P_k^- - K_kS_kK_k^T, \end{aligned} \quad (2.15)$$

the Jacobian matrices $F_x(m_k^-)$ and $H_x(m_k^-)$ are the Taylor series linear approximation of the dynamic and measurement models

$$\begin{aligned} [F_x(m)]_{j,j'} &= \left. \frac{\partial f_j(x)}{\partial x_{j'}} \right|_{x=m_{k-1}} \\ [H_x(m)]_{j,j'} &= \left. \frac{\partial h_j(x)}{\partial x_{j'}} \right|_{x=m_k^-}. \end{aligned} \quad (2.16)$$

The intrinsic bottleneck of the extended Kalman filter can be ascribed to the Gaussian formulation of the state update equations. Any deviations from Gaussianity in the posterior will directly degrade the estimation capabilities of the estimator. Another type of state estimation method is the particle filter, which can accommodate nonlinear and nongaussian dynamic and measurement models by using point estimates from Monte Carlo approximation of the posterior.

2.1.2 Bootstrap particle filter

When revisiting the probabilistic state estimators in section 2.1, to analytically compute the point estimate of the states $\hat{x}$, given all observations up to time step K , one must evaluate the expected value of the conditional probability

$$\hat{x} = E[g(x) \mid y_{1:K}] = \sum_x g(x)p(x \mid y_{1:K}), \quad (2.17)$$

with $g(\cdot)$ being an arbitrary function $\mathbb{R}^n \rightarrow \mathbb{R}^m$. Such an integral is often difficult to compute in closed-form due to the intractability of the posterior $p(x \mid y_{1:K})$ in many settings, but by formulating a Monte Carlo approximation by drawing N many random samples from the target distribution $x^{(i)} \sim p(x \mid y_{1:K})$ $i = 1, \dots, N$, the approximation of the point estimate of the states will converge to the true point estimate for a large N , according to the Law of Large Numbers and the Central Limit Theorem [21], and the approximation of the point estimate of the states can be rewritten as

$$\hat{x} \approx \frac{1}{N} \sum_{i=1}^N g(x^{(i)})p(x^{(i)} \mid y_{1:K}). \quad (2.18)$$

The conditional probability $p(x^{(i)} \mid y_{1:K})$ can be rewritten in terms of the likelihood $p(y_{1:K} \mid x^{(i)})$ and the prior $p(x^{(i)})$ according to Bayes' rule

$$p(x^{(i)} \mid y_{1:K}) = \frac{p(y_{1:K} \mid x^{(i)})p(x^{(i)})}{p(y_{1:K})}. \quad (2.19)$$

The normalization term is often not able to be computed numerically, but by introducing importance sampling, the term can be approximated. The importance mass $\pi(x^{(i)} \mid y_{1:K})$ is a Monte Carlo approximation of $p(y_{1:K})$, with the condition that the mass is non-zero whenever $p(x^{(i)} \mid y_{1:K})$ is non-zero, which results in a nonuniform distribution. Sampling from a nonuniform distribution will result in each particle having a corresponding weight $w^{*(i)}$, and a normalized weight w^i

$$\begin{aligned} w^{*(i)} &= \frac{p(y_{1:K} \mid x^{(i)})p(x^{(i)})}{\pi(x^{(i)} \mid y_{1:K})} \\ w^i &= \frac{w^{*(i)}}{\sum_{j=1}^N w^{*(j)}}, \end{aligned} \quad (2.20)$$

with the approximation of the point estimate being rewritten as

$$\hat{x} \approx \sum_{i=1}^N w^i g(x^{(i)}). \quad (2.21)$$

The bootstrap particle filter is formulated on the basis of Markov Chain Monte Carlo approximations, the filtering distributions are shown in Equation (2.9), and the filter is run sequentially like the Kalman filter, the importance mass is assigned as the dynamic model $p(x_k | x_{k-1}^{(i)})$ at the time step k for a given particle (i) , which results in the following algorithm [20] [21]

1. Sample N particles from the dynamic function

$$x_k^{(i)} = p(x_k | x_{k-1}^{(i)}, u_{k-1}) \quad i = 1, \dots, N \quad (2.22)$$

2. Update weights from the measurement

$$\begin{aligned} w_k^{*(i)} &\propto p(y_k | x_k^{(i)}) \quad i = 1, \dots, N \\ w_k^{(i)} &= \frac{w_k^{*(i)}}{\sum_{j=1}^N w_k^{*(j)}} \quad i = 1, \dots, N \end{aligned} \quad (2.23)$$

3. Resample particles from a distribution proportional to the weights.

2.1.3 Observability and Identifiability

Analyzing observability and identifiability in linear and nonlinear systems assesses how well the dynamics of the model can be inferred from the observations. The observability gramian W_O has traditionally been used in model reduction of linear dynamic systems, but empirical methods have been adapted to accommodate nonlinear systems [22]. The observability gramian provides insight into which states or parameters have the most influence on the output of the system by analyzing the rank condition of the observability gramian matrix, which in return allows for analytically evaluating observability and identifiability.

We can define the following generic discrete system with a nonlinear dynamic function $f(\cdot)$ and a nonlinear measurement function $h(\cdot)$

$$\begin{aligned} \dot{x}_k &= f(x_{k-1}, u_k) \\ y_k &= h(x_k). \end{aligned} \quad (2.24)$$

To empirically calculate the observability gramian, we first generate 2^n unique combinations of the initial state $x_0 \in \mathbb{R}^n$, with n denoting the numbers of states and parameters. The initial condition x_0 will be assigned as perturbations $\delta_i \in \mathbb{R}^n$ around a nominal value $x_{nom} \in \mathbb{R}^n$

$$\begin{aligned} x_{0,i} &= x_{nom} + \delta_i \\ \delta_i &= c \cdot S \cdot t_i, \quad \text{with } S = \text{diag}(x_{\max}). \end{aligned} \quad (2.25)$$

The scalar constant c corresponds to the magnitude of deviation, $S \in \mathbb{R}^{n,n}$ is the diagonal matrix of the maximum of each state and parameter. The row vector $t_i \in \mathbb{R}^n$ is the specified perturbation i from the orthonormal matrix $T = [t_1 \ t_2 \ \dots \ t_{2^n}] \in \mathbb{R}^{n,2^n}$

$$\begin{aligned} t_1^T &= \frac{1}{\sqrt{2^n}} [-1 \ -1 \ \dots \ -1] \\ t_2^T &= \frac{1}{\sqrt{2^n}} [-1 \ -1 \ \dots \ 1] \\ &\vdots \\ t_{2^n}^T &= \frac{1}{\sqrt{2^n}} [1 \ 1 \ \dots \ 1]. \end{aligned} \quad (2.26)$$

After calculating the 2^n perturbations of the initial state, the output can be derived iteratively with Equations (2.24) from time step $k = 1, \dots, K$ beginning with $x_{0,i}$ to generate $y_i [1 : K]$. Then, the empirical observability with discrete time steps $t = k\Delta T$ can be expressed as such

$$W_O = (cS)^{-1} \sum_{k=1}^K T \Psi[k] T^T \Delta T (cS)^{-1}, \quad (2.27)$$

with the positive definite matrix $\Psi[k]$ being assigned as the following

$$\Psi_{i,j}[k] = |y_i[k] - y_0[k]| |y_j[k] - y_0[k]|. \quad (2.28)$$

To make inferences about the observability and identifiability of the observability gramian, one can check the rank condition of the matrix by analyzing the nonzero singular values of W_0 . The singular values close to zero can be classified as not observable. If the states have varying magnitudes of scale, it is recommended [18] to rerun the observability gramian algorithm again with the following transformations

$$x_s = \sqrt{\text{diag}(W_O)} \quad T_s = \text{diag}^{-1}(x_s). \quad (2.29)$$

The new state perturbations are now assigned as the transformation $\bar{x} = T_s x$. If the system is found to be weakly observable or not observable after calculating the singular values with the transformed parameters, consequently, a subset of the dynamics of the system can't be observed from the measurements. State estimators will be discussed in the following sections, which can account for the lack of intuition into the dynamics not observable through various probabilistic filtering mechanisms.

2.2 Parameter Estimation

As previously mentioned, there exists patient-specific parameters. The whole-body insulin sensitivity S_I or the endogenous insulin u_{en} dynamically fluctuates over time depending on the patient's metabolic condition. As a result, naively assigning either a population constant can hinder the performance of the estimators. The unknown constant parameters of interest will be denoted as θ , with different proposed computational methods to estimate them will be elaborated on within this section.

2.2.1 State Augmentation

A practical method to handle unknown parameters is to augment them to the existing state variables x_k , such that Equation (2.9) can be redefined as

$$\begin{aligned} \theta_k &= \theta_{k-1} \\ x_k &\sim p(x_k \mid x_{k-1}, u_{k-1}, \theta_{k-1}) \\ y_k &\sim p(y_k \mid x_k). \end{aligned} \quad (2.30)$$

An important distinction in Equation (2.30) is that the unknown parameters are treated as stochastic variables with singularity in the process noise. This can be problematic as nonlinear models often have approximation errors, and

this can drive the extended Kalman filter to ultimately diverge in the Gaussian approximation, as well as introduce sample impoverishment in particle filters [20].

One method to compensate for singularity is to add artificial process noise Q_k to allow the unknown parameters to randomly walk in the state space, thus, accounting for errors in the dynamic function

$$\theta_k = \theta_{k-1} + \epsilon_{k-1}, \quad \text{with } \epsilon_{k-1} \sim N(0, Q_{k-1}). \quad (2.31)$$

With this method, you inherently add another unknown parameter Q_k , but such a parameter can be tuned with prior knowledge of the parameters θ_k , or one can utilize the Rao-Blackwellized representation of the covariance Q_k in the state space.

2.2.2 Markov Chain Monte Carlo Sampling

The Markov Chain Monte Carlo methods are a class of parameter estimation approaches to estimate the posterior $p(\theta \mid y_{1:K})$ by drawing random samples of the unknown parameters θ from a desired stationary distribution formulated as a Markov Chain [20][23]. Only the unnormalized posterior is considered due to similar concerns regarding computing $p(y_{1:K})$ as previously discussed in Section 2.1.2

$$p(\theta \mid y_{1:K}) \propto p(y_{1:K} \mid \theta)p(\theta). \quad (2.32)$$

The marginal likelihood $p(y_{1:K} \mid \theta)$ can be represented recursively as the state space models are run sequentially

$$p(y_{1:K} \mid \theta) = \prod_{k=1}^K p(y_k \mid y_{k-1}, \theta). \quad (2.33)$$

The energy function $\varphi(\theta)$ will be introduced to maximize $p(y_{1:K} \mid \theta)$ in the negative log-domain

$$\varphi(\theta) = -\log p(y_{1:K} \mid \theta) = -\sum_{k=1}^K \log p(y_k \mid y_{k-1}, \theta), \quad (2.34)$$

which can additionally be expressed as in the recursive form

$$\varphi_k(\theta) = \varphi_{k-1}(\theta) - \log p(y_k \mid y_{k-1}, \theta). \quad (2.35)$$

2.2.2.1 Extended Kalman Filter-based Energy Function

If we return to Equations (2.14) and (2.15), the equations will be redefined such that the filter parameters are also dependent on the unknown parameters θ

1. Prediction step

$$\begin{aligned} m_k^-(\theta) &= f(m_{k-1}(\theta), u_{k-1}) \\ P_k^-(\theta) &= F_x(m_{k-1}(\theta))P_{k-1}(\theta)F_x^T(m_{k-1}(\theta)) + Q_{k-1}(\theta) \end{aligned} \quad (2.36)$$

2. Update step

$$\begin{aligned} v_k(\theta) &= y_k - h(m_k^-(\theta)) \\ S_k(\theta) &= H_x(m_k^-(\theta))P_k^-(\theta)H_x^T(m_k^-(\theta)) + R_k(\theta) \\ K_k(\theta) &= P_k^-(\theta)H_x^T(m_k^-(\theta))S_k^{-1}(\theta) \\ m_k(\theta) &= m_k^-(\theta) + K_k(\theta)v_k(\theta) \\ P_k(\theta) &= P_k^-(\theta) - K_k(\theta)S_k(\theta)K_k^T(\theta). \end{aligned} \quad (2.37)$$

The marginal log likelihood $\log p(y_k \mid y_{k-1}, \theta)$ in the extended Kalman filter can be simply derived as

$$\begin{aligned} \log p(y_k \mid y_{k-1}, \theta) &= \log N(y_k \mid h(m_k^-(\theta)), S_k(\theta)) \\ &= \log \frac{1}{\sqrt{|2\pi S_k(\theta)|}} \exp \left(-\frac{1}{2} v_k^T(\theta) S_k(\theta) v_k(\theta) \right) \\ &= -\frac{1}{2} \log |2\pi S_k(\theta)| - \frac{1}{2} v_k^T(\theta) S_k(\theta) v_k(\theta). \end{aligned} \quad (2.38)$$

Given the closed form formulation of the marginal log likelihood, the energy function in Equation (2.35) can be finalized as

$$\varphi_k(\theta) = \varphi_{k-1}(\theta) + \frac{1}{2} \log |2\pi S_k(\theta)| + \frac{1}{2} v_k^T(\theta) S_k(\theta) v_k(\theta). \quad (2.39)$$

2.2.3 Metropolis-Hastings Algorithm

The Metropolis-Hastings algorithm is one of the most common Markov Chain Monte Carlo samplers, and provides a sequential process of estimating the potential candidates of θ by drawing them from a symmetric proposal distribution $\theta^* \sim q(\theta^* \mid \theta_{(k-1)})$, given the previous candidate $\theta_{(k-1)}$ [20] [23]. The algorithm will be laid out in the preceding block

- $k = 0$
 1. assign an initial candidate θ_0 and the initial energy function value $\varphi_0(\theta_0) = -\log p(\theta_0)$
- for $k = 1 : K$
 1. sample a proposal candidate $\theta^* \sim q(\theta^* | \theta_{k-1})$
 2. calculate energy of proposed candidate $\varphi_k(\theta^*)$
 3. calculate acceptance probability $\alpha_k = \min(1, \exp(\varphi_k(\theta^*) - \varphi_k(\theta_{k-1})))$
 4. $\theta_k = \begin{cases} \theta^* & \text{if } z < \alpha_k \\ \theta_{k-1} & \text{otherwise} \end{cases}$ with $z \sim \mathcal{U}[0, 1]$

2.3 Related Work

A nonlinear, and nonconvex integral-based parameter identification method has been introduced that can estimate patient-specific parameters through maximum likelihood estimation [15]. By linearly interpolating between the blood glucose measurements in times $[t_0, t_1]$,

$$\widetilde{G(t)} = \frac{G(t_1) - G(t_0)}{t_1 - t_0}t, \quad (2.40)$$

the Equation (2.1) can be rearranged as such for a piecewise constant S_I

$$\begin{aligned} \int_{t_0}^{t_1} \dot{G} dt &= -p_G \int_{t_0}^{t_1} \widetilde{G(t)} dt - S_I \int_{t_0}^{t_1} \widetilde{G(t)} \frac{Q(t)}{1 + \alpha_G Q(t)} dt \\ &\quad + \frac{1}{V_G} \int_{t_0}^{t_1} P(t) + E_G - C_G dt \\ G(t_1) - G(t_0) &= -p_G \int_{t_0}^{t_1} \widetilde{G(t)} dt - S_I \int_{t_0}^{t_1} \widetilde{G(t)} \frac{Q(t)}{1 + \alpha_G Q(t)} dt \\ &\quad + \frac{1}{V_G} \int_{t_0}^{t_1} P(t) + E_G - C_G dt, \end{aligned} \quad (2.41)$$

and S_I can be expressed as a linear equation

$$\underbrace{\int_{t_0}^{t_1} \widetilde{G}(t) \frac{Q(t)}{1 + \alpha_G Q(t)} dt}_{A(t_0, t_1)} S_I = \underbrace{\left(G(t_0) - G(t_1) - p_G \int_{t_0}^{t_1} \widetilde{G}(t) dt + \frac{1}{V_G} \int_{t_0}^{t_1} (P(t) + E_G - C_G) dt \right)}_{b(t_0, t_1)}. \quad (2.42)$$

It is often the case with longer measurement periods ($t_1 - t_0$), that identifying multiple S_I values within the time period may be necessary. For instance, if every 120 mins glucose measurements are taken from the patient, but every 60 mins a patient's insulin sensitivity should be identified $S_I \in \mathbb{R}^2$. Both insulin sensitivity parameters at 60 mins S_I^{60} and 120 mins S_I^{120} can be derived as the least squares solution from the system of linear equations

$$\begin{bmatrix} A(t_0, t_0 + 30) & 0 \\ A(t_0 + 30, t_0 + 60) & 0 \\ 0 & A(t_0 + 60, t_0 + 90) \\ 0 & A(t_0 + 90, t_0 + 120) \end{bmatrix} \begin{bmatrix} S_I^{60} \\ S_I^{120} \end{bmatrix} = \begin{bmatrix} b(t_0, t_0 + 30) \\ b(t_0 + 30, t_0 + 60) \\ b(t_0 + 60, t_0 + 90) \\ b(t_0 + 90, t_0 + 120) \end{bmatrix}. \quad (2.43)$$

The fundamental flaw of the integral-based parameter identification method is the assumption of linear blood glucose dynamics between measurements of times $[t_0, t_1]$. If we closely analyze the Figure 2.1, linearly interpolating blood glucose introduces residuals, due to fitting errors. Consequently, the residuals will propagate into the integrals in the Equation (2.42), resulting in errors in the parameter calculation. Relying on data-driven methods can not account for physiological behavior in the parameters, and other methods must be explored to allow for realistic estimation of parameters, thus giving better insight of the patient's health status to clinicians.

*The initial conditions of the patient model are $BG = 5.0$ (mmol/L), $I = 150$ (mU/L), $Q = 100.0$ (mU/L), $P1 = 7$ (mmol), $P2 = 32$ (mmol). The inputs are $u_{ex}(t) = 125 \times e^{-\left(\frac{\log(2)}{300}\right) \cdot ((t+120) \bmod (300))}$ (mU/min), $P_N(t) = e^{-\left(\frac{\log(2)}{300}\right) \cdot (t \bmod (300))}$ (mmol/min), $D(t) = 0.200$ (mmol/min). Measurements are sampled with additive Gaussian noise $r_k \sim N(0, 0.25^2)$.

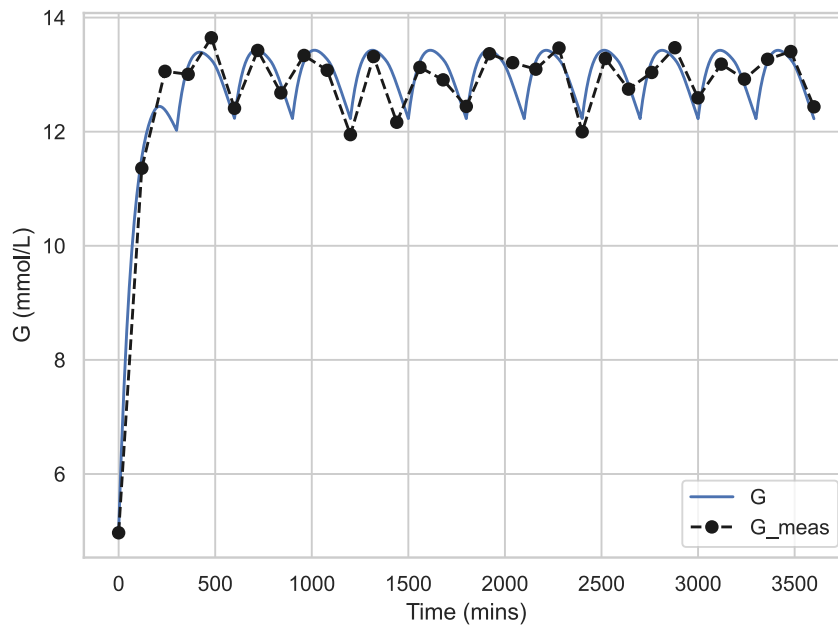


Figure 2.1: Blood glucose and its' corresponding measurements from the patient model*.

2.4 Summary

The Intensive Control Insulin-Nutrition-Glucose model provides a foundation to model glucose removal, insulin pharmacokinetics, and gastric absorption of glucose for tight glucose control in critically ill patients. Often, clinicians solely rely on blood glucose measurements, which limits the knowledge of the insulin-glucose dynamics in real-time. Utilizing nonlinear state estimators, such as the extended Kalman filter and the bootstrap particle filter, allow for estimation of all dynamics over time. To evaluate patient-specific parameters, model identifiability is used to make inferences about the parameters of interest. The current method of integral-based parameter identification is formulated with strict assumptions of blood glucose measurements, which limits the ability for clinicians to infer the health of the patient from the approximated patient-specific parameters. More in-depth parameter estimation methods can improve model identifiability for glucoregulation in critically ill patients.

Chapter 3

Methodology

This chapter will consist of presenting the scientific design of the experiment. Beginning with the overall goals of the study, and how the corresponding experimental process will be formulated to achieve answering the stated research questions. The inclusion and motivation of the data collected, models used, and analysis of the observations in the experiment will be laid out in the following sections.

3.1 Research Process

The main topics this study aim to address fall in line with the following research questions:

1. **Does the mathematical formulation of the Intensive Control Insulin-Nutrition-Glucose model permit patient-specific model-identifiability when approximating insulin-glucose dynamics for patients in the intensive care unit?**
2. **How do state estimation techniques compare against traditional parameter-identification methods in estimating patient-specific insulin-glucose dynamics in terms of accuracy and computational complexity?**

Given the overall experimental goals, the research process can therefore be outlined as such:

Step 1 create virtual patients,

Step 2 conduct experiments with patient data,

Step 3 analyze statistical metrics from the experiments, and

Step 4 discuss the results of the analysis.

3.2 Research Paradigm

The philosophical motivation in this study is founded on utilizing a set of quantitative methods that serve the purpose of ultimately answering the previously stated research questions. The use of statistical metrics will quantitatively express the relationships of variables tested in the experiment. Consequently, the research questions will effectively be addressed.

3.3 Data Collection

A virtual patient cohort will be simulated to model clinical insulin-glucose data observed in patients in the intensive care unit. A patient model consists of simulated insulin-glucose dynamics ($G, Q, I, P1, P2, u_{en}, P$) represented in Equations (2.1) to (2.7), known insulin sensitivity $S_I(t)$, exogenous insulin $u_{ex}(t)$, enteral feed $D(t)$, and parenteral feed $PN(t)$. The population constants listed in Table 2.1 are assigned to the constant parameters for all virtual patients. To introduce variance between patients, a virtual patient is given a random constant input observed from doses given to patients in the Karolinska University Hospital

$$\begin{aligned} u_{ex} &\sim \mathcal{U}[0, 166] \text{ mU/min} \\ D &\sim \mathcal{U}[0.2, 0.4] \text{ mmol/min} \\ PN &\sim \mathcal{U}[0.2, 0.4] \text{ mmol/min.} \end{aligned} \tag{3.1}$$

The insulin sensitivity SI has been shown to follow a log-normal distribution [24]. Sampling the insulin sensitivity from a log-normal is important to adhere to the physiological constraints typically observed in patients, as the insulin sensitivity is small in magnitude, and sampling from a normal distribution may shift the value to negative or too large in magnitude, which can possibly result in poor estimation. The values chosen within this study will be drawn from the distribution

$$SI = \exp(X), \quad \text{with } X \sim N(-8.7, 0.40^2). \tag{3.2}$$

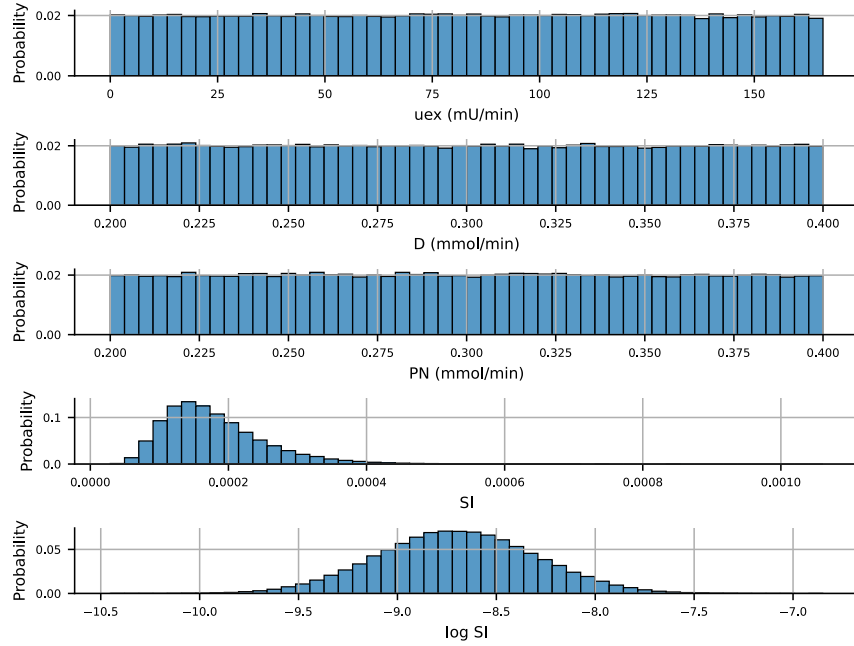


Figure 3.1: Probability Density Functions of the system inputs and parameters.

To simulate patient data given the inputs and parameters, each virtual patient needs to be assigned an unique initial state for the dynamics

$$x_0 = [G_0 \quad I_0 \quad Q_0 \quad P1_0 \quad P2_0 \quad P(0), uen(0)] . \quad (3.3)$$

If it assumed that the patient model starts at steady state, it is possible to derive the initial states with the input and parameter values by solving for the roots of the Differential Equations (2.1) to (2.6). This can be computed with the solver function `scipy.optimize.fsolve*` for each patient.

Given all the initial states x_0 , known insulin sensitivity $S_I(t)$, and the patient inputs $u_{ex}(t)$, $D(t)$, the patient cohort can be generated by solving the Differential Equations (2.1) to (2.5) with a time step of $dt = 1$ min for each patient, and the results will be inputs to Equations (2.6) and (2.7). The ODE solver `scipy.integrate.solve_ivp†` function in python will be applied with the Explicit Runge-Kutta of order 5(4) integration method [25] to solve for the necessary dynamics.

*<https://docs.scipy.org/doc/scipy/reference/generated/scipy.optimize.fsolve.html>

†https://docs.scipy.org/doc/scipy/reference/generated/scipy.integrate.solve_ivp.html

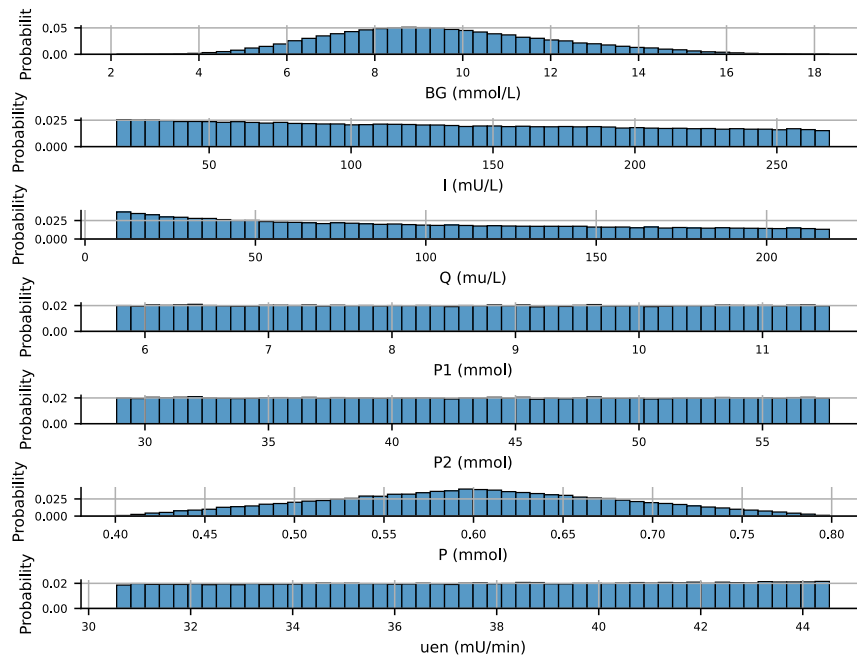


Figure 3.2: Probability Density Function of the initial states.

3.4 Assessing reliability and validity of the data collected

3.4.1 Validity of method

The patient models, parameters, and initial states are all adopted and cross-validated from multiple clinically validated studies. This provides a physiologically accurate basis for the control variables of the experiment. Interventions in the experiment can be assessed and compared to the manipulated variables, and with confidence a conclusion can be reached that the intervention is correlated with the change in the output, and not due to poor model formulation with nonvalidated patient input and parameters.

3.4.2 Reliability of method

Simulating patient data gives the necessary control to assign all patient parameters. A clear benefit of virtual patients is the repeatability of experiments, given identical patient parameters. Often in clinical settings, the true patient dynamics and parameters are unknown. Thus, results from

clinical data may not be repeatable under the same conditions as there could be unaccounted for variables in the methods.

3.5 Evaluation framework

To answer the given Research Questions 1 and 2, statistical metrics such as Mean Square Error and Mean Absolute Relative Error, and mathematical results (e.g. identifiability and time-complexity of estimators) will be used to quantify the results of the experiment such that the hypotheses can be tested.

In order to evaluate the estimation of patient-specific parameters, the methodology will follow a trajectory of investigating patient identifiability in insulin-glucose models, comparing state estimators in stochastic models, and then assessing parameter estimation methods. Each phase of the experiment will have their own respective patient cohort, and statistical metrics. As previously mentioned, measurements of the patient model y_k correspond on lone observations of blood glucose corrupted with zero-mean Gaussian noise, which will be used for all of the following sections.

3.6 Observability and Identifiability in Patient Insulin-Glucose Models

In Chapter ??, a patient model was created for each individual patients based on the Intensive Control Insulin-Glucose model, with the ensemble of patient models being labeled as a patient cohort. For a particular patient model to be identifiable, it must be possible to learn the underlying unique patient-specific parameters that define the model, given a sufficient number of observations. To establish if the formulation of the patient model is identifiable, a mathematical analysis can determine the relationship of the insulin-glucose dynamics and the observations as previously discussed in Section 2.1.3.

1000 patient models will be randomly sampled to investigate the observability of the Intensive Control Insulin-Nutrition-Glucose model across various permutations of the nominal state vector*. The maximum value of

*While u_{en} is a process variable, the numerical integration method solves for the Differential Equations (2.1) to (2.5). Thus, permutations in u_{en} do not result in a change in G . For the observability gramian calculation, u_{en} is omitted from x_0 .

each state x_{max} will be chosen from Figure 3.2 to assign S

$$\begin{aligned} S &= \text{diag}(x_{max}) \\ &= \text{diag}([17.5 \quad 260 \quad 220 \quad 11.5 \quad 57]), \end{aligned} \quad (3.4)$$

as well with a maximum perturbation constant of $c = 0.1$. With the following defined inputs, the observability gramian W_O can be derived to analyze the singular values of all the calculated matrices.

3.7 Nonlinear State Estimation

The extended Kalman filter and the particle filter are two different nonlinear state estimation methods presented in Section 2.1. The key distinctions between the two estimators can be summarized as particle filters rely on Monte Carlo simulations of each state and the particles are resampled after each measurement, while the extended Kalman filter is formulated on a linear approximation of the state-space transformations with Gaussian distributed states and measurements. To compare each estimator with the Intensive Control Insulin-Nutrition-Glucose model, the total time to compute all the estimated states and the squared error of the blood glucose estimates $(G[k] - \hat{G}[k])^2$ will be recorded for all patients in the given virtual patient cohort. The simulations will span 6 hours with a time-step of 1 minute. There will be iterations of this comparison with the following offsets of the initial states: $\pm 5\%$, $\pm 10\%$, $\pm 15\%$, $\pm 20\%$, with each iteration having a virtual cohort of 500 patients. Adding an offset to the initial state will gauge how each well estimator can adapt to various initial innovations in the estimate.

The number of particles for the particle filter is a hyperparameter that will be analyzed within this section. Particle filters containing (100, 200, 300, 400, 500, 750, 1000, 1500, 2000) particles will be used to compare the relationship of the outputs over an increasing number of particles. A Pareto curve can be formulated to represent the "optimal" number of particles in terms of the trade off between time complexity and error. The resulting number of particles needed per dimension will be calculated to gauge the rough number of particles needed to include additional parameters in the state-space in the following section.

3.8 Parameter Estimation

The Integral-based Parameter Identification method introduced in Section 2.3 will be the baseline implementation of estimation patient-specific parameters.

Chapter 4

Results and Analysis

4.1 Major results

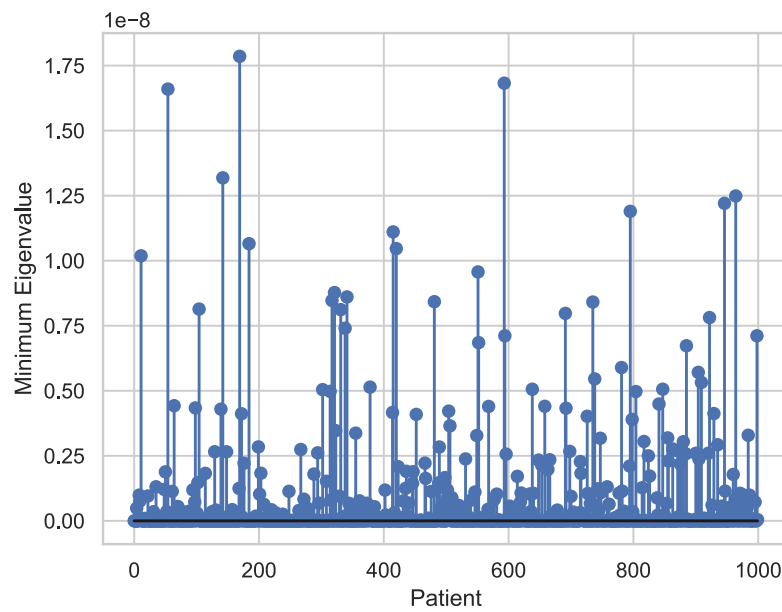


Figure 4.1: Minimum eigenvalues from the corresponding observability gramian across patients.

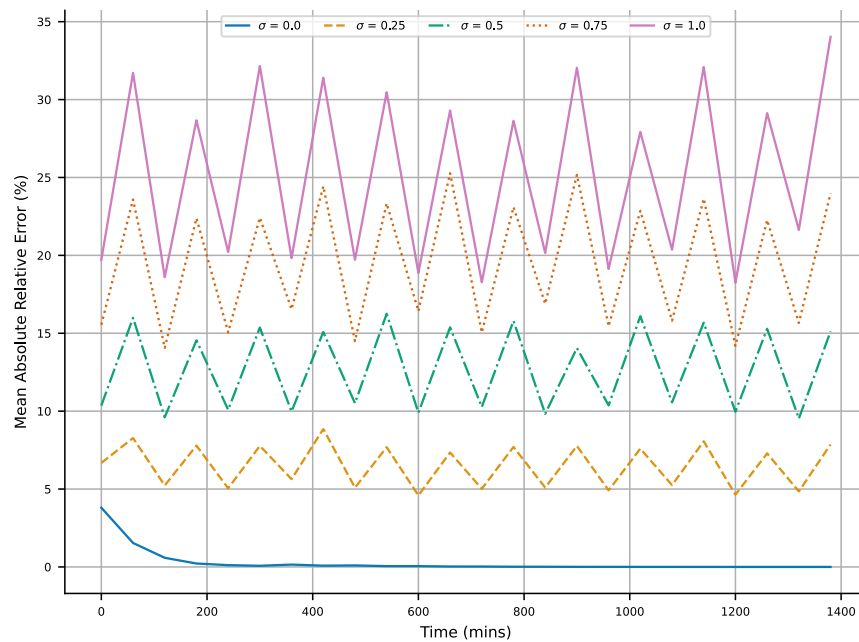


Figure 4.2: Mean Absolute Relative Error of insulin sensitivity estimates over increasing Gaussian measurement noise magnitudes with the Integral-based Parameter Identification method.

4.2 Reliability Analysis

Analys av tillförlitlighet
Tillförlitlighet i metod och data

4.3 Validity Analysis

Analys av validitet
Validitet i metod och data

Chapter 5

Discussion

Chapter 6

Conclusions and Future work

Slutsats och framtida arbete

Add text to introduce the subsections of this chapter.

6.1 Conclusions

Slutsatser

Describe the conclusions (reflect on the whole introduction given in Chapter 1).

Discuss the positive effects and the drawbacks.

Describe the evaluation of the results of the degree project.

Did you meet your goals?

What insights have you gained?

What suggestions can you give to others working in this area?

If you had it to do again, what would you have done differently?

Uppfyllde du dina mål?

Vilka insikter har du fått?

Vilka förslag kan du ge till andra som arbetar inom detta område? Om du skulle göra detta igen, vad skulle du ha gjort annorlunda?

6.2 Limitations

Begränsande faktorer

Vad gjorde du som begränsade dina ansträngningar? Vilka är begränsningarna i dina resultat?

What did you find that limited your efforts? What are the limitations of your results?

6.3 Future work

Vad du har kvar ogjort?

Vad är nästa självklara saker som ska göras?

Vad tips kan du ge till nästa person som kommer att följa upp på ditt arbete?

Describe valid future work that you or someone else could or should do. Consider: What you have left undone? What are the next obvious things to be done? What hints can you give to the next person who is going to follow up on your work?

Due to the breadth of the problem, only some of the initial goals have been met. In these section we will focus on some of the remaining issues that should be addressed in future work. ...

6.3.1 What has been left undone?

The prototype does not address the third requirement, *i.e.*, a yearly unavailability of less than 3 minutes, this remains an open problem. ...

6.3.1.1 Cost analysis

Example of a missing component

The current prototype works, but the performance from a cost perspective makes this an impractical solution. Future work must reduce the cost of this solution, to do so a cost analysis needs to first be done. ...

6.3.1.2 Security

Example of a missing component

A future research effort is needed to address the security holes that results from using a self-signed certificate. Page filling text mass. Page filling text mass. ...

6.3.2 Next obvious things to be done

In particular, the author of this thesis wishes to point out xxxxxx remains as a problem to be solved. Solving this problem is the next thing that should be done. ...

6.4 Reflections

Reflektioner

Vilka är de relevanta ekonomiska, sociala, miljömässiga och etiska aspekter av ditt arbete?

What are the relevant economic, social, environmental, and ethical aspects of your work?

One of the most important results is the reduction in the amount of energy required to process each packet while at the same time reducing the time required to process each packet.

The thesis contributes to the **United Nations (UN) Sustainable Development Goals (SDGs)** numbers 1 and 9 by xxxx.

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Appendix A

Something Extra

svensk: Extra Material som Bilaga

A.1 Just for testing KTH colors

You have selected to optimize for print output

- Primary color

- kth-blue 

- kth-blue80 

- Secondary colors

- kth-lightblue 

- kth-lightred 

- kth-lightred80 

- kth-lightgreen 

- kth-coolgray 

- kth-coolgray80 

black 

Appendix B

Main equations

This appendix gives some examples of equations that are used throughout this thesis.

B.1 A simple example

The following example is adapted from Figure 1 of the documentation for the package `nomencl` (<https://ctan.org/pkg/nomencl>).

$$a = \frac{N}{A} \tag{B.1}$$

The equation $\sigma = ma$ follows easily from Equation (B.1).

B.2 An even simpler example

The formula for the diameter of a circle is shown in Equation (B.2) area of a circle is shown in eq. (B.3).

$$D_{circle} = 2\pi r \tag{B.2}$$

$$A_{circle} = \pi r^2 \tag{B.3}$$

Some more text that refers to (B.3).

€€€€ For DIVA €€€€

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acronyms.tex

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%%% mode: latex
%%% TeX-master: t
%%% End:
% The following command is used with glossaries-extra
\setabbreviationstyle{acronym}{long-short}
% The form of the entries in this file is \newacronym{label}{acronym}{phrase}
%                                     or \newacronym[options]{label}{acronym}{phrase}
% see "User Manual for glossaries.sty" for the details about the options, one example is shown below
% note the specification of the long form plural in the line below
\newacronym[longplural={Debugging Information Entities}]{DIE}{DIE}{Debugging Information Entity}
%
% The following example also uses options
\newacronym[shortplural={OSes}, firstplural={operating systems (OSes)}]{OS}{OS}{operating system}

% note the use of a non-breaking dash in long text for the following acronym
\newacronym{IQL}{IQL}{Independent Q-Learning} % <-- regular dash

\newacronym{KTH}{KTH}{KTH Royal Institute of Technology}

\newacronym{LAN}{LAN}{Local Area Network}
\newacronym{VM}{VM}{virtual machine}
% note the use of a non-breaking dash in the following acronym
\newacronym{WiFi}{Wi-Fi}{Wireless Fidelity}

\newacronym{WLAN}{WLAN}{Wireless Local Area Network}
\newacronym{UN}{UN}{United Nations}
\newacronym{SDG}{SDG}{Sustainable Development Goal}
```